Jaeyong Ko

M.S. Student
Industrial Engineering
Seoul National University
Data Science & Business Analytics Lab

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Homepage
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Research Interests

- **Current Focus:** Visual Reasoning, Interleaved Vision-Language Reasoning, Latent Reasoning
- **Previous Research:** LLM Mathematical Reasoning, LLM Evaluation in the Financial Domain

Education

- **M.S.** *Industrial Engineering* — Sep. 2025 – Present
  Seoul National University — *Advisor: Prof. Pilsung Kang*
- **B.S.** *Industrial Engineering* — Mar. 2019 – Aug. 2025
  Seoul National University — *GPA: 3.72/4.0*

Publications

Oddin: Latent Visual Reasoning in Unified Multimodal Models via Self-Distillation from Image Generation 2026
Jaeyong Ko*, Hankyeol Kim*, Pilsung Kang.
arXiv preprint arXiv:2610.33062. [arXiv]

FinLifeBench: Exhaustive Life-Event History and Financial-State Reconstruction from Longitudinal Banking Dialogue 2026
Hangyeul Lee*, Juyoung Oh*, **Jaeyong Ko**, Sunmin Kim, Jaeik Park, Hyunkyu Kim, Jungmin Son, Pilsung Kang†.
ACM ICAIF 2026 (Accepted). [arXiv]

Cliff Tokens: Analyzing Failure Trigger Tokens in LLM Mathematical Reasoning 2026
Jaeyong Ko, Jinu Lee, Pilsung Kang, Yukyung Lee†.
arXiv preprint arXiv:2606.25524. [project page] [arXiv] [github] [dataset]

Educational Publications

Can a model think VISUALLY? 2026
Jaeyong Ko, Eunseo Kim, Doehyeon Lee.
NeurIPS 2026 Education Track (Accepted).

AI Safety Lives in a Direction: Reading, Breaking, and Repairing Refusal in Multimodal Models 2026
Doehyeon Lee, Yoo-Min Jung, **Jaeyong Ko**.
NeurIPS 2026 Education Track (Accepted).

Research Experience

- **Research Lead** *Visual Latent Reasoning for Unified Models* — Jul. 2026 – Present
  Transferring visual reasoning capabilities from the generation branch of unified models to the understanding branch
- **Research Intern** *École Polytechnique de Montréal* — Apr. 2024 – Aug. 2024
  Human-centric Smart Manufacturing Workcells for Industry 5.0 applications. — *Advisor: Prof. Walid Jomaa*
- **Research Intern** *Financial Engineering Lab, KAIST* — Dec. 2023 – Feb. 2024
  Reinforcement learning applications for goal-based wealth management. — *Advisor: Prof. Woo Chang Kim*

Projects

- **Development of Core Foundation Model and AI Agent Technology for Thermal-Process Manufacturing** | *Granted by Ministry of Trade, Industry and Energy* — Sep. 2026 – Present
  - Designed the backbone architecture for the thermal-process foundation model and established an integrated learning framework utilizing domain data and PINN.
- **Duo 2.0: Evaluating LLM Reliability in Long Financial Conversations** | *Granted by KakaoBank* — Dec. 2025 – Present
  - Quantified LLM capabilities in resolving ambiguous user queries using LLM-as-a-Judge methodologies.
  - Designed a financial life-cycle event conversation benchmark for domain-specific conversational reliability.

- **Development of Advanced Integrated Monitoring Technologies for Laser Welding** Jul. 2025 – Dec. 2025
  | *Granted by LG Electronics*
  - Designed and conducted experiments for a laser welding anomaly detection model utilizing time-series and image multimodal data.

Industry Experience

- **President & Team Member** *SNU Startup Consulting Group (ACT)* Mar. 2023 – Dec. 2023
  - Led a student consulting group and secured four startup consulting engagements with *Moderntec*, *Designovel*, *Spring Onward*, and *Sweetspot*.
  - Developed GTM strategies for *Designovel* (AI fashion-tech), *Moderntec* (EV charging infrastructure), and *Wavebridge* (cryptocurrency OTC platform).
- **Product Manager Intern** *EXGRAM* Jul. 2022 – Oct. 2022
  - Planned app configurations and push notifications for the app, Veloo.

Teaching Experience

- **Peer Tutor** *Production Control (406.310)* Sep. 2024 – Dec. 2024
  Seoul National University
- **Peer Tutor** *Human Factors Engineering (406.304)* Sep. 2023 – Dec. 2023
  Seoul National University

Honors & Awards

- **Excellence Award** *Poster Presentation* Aug. 2026
  2026 KDMS Summer Conference, Korea Data Mining Society

Skills

- **Programming and Software:** Python, R, MySQL, Docker, Git, LaTeX.
- **Languages:** Native Korean; professional working proficiency in English (TOEFL 5.0/6.0, OPIc IH).

Military Service

- **Corporal** *Republic of Korea Army* Jan. 2021 – Jul. 2022
  Completed full-term mandatory military service.